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PROFESSIONAL SUMMARY

- Data Scientist with over 6 years of experience delivering predictive analytics, statistical modeling, data science workflows and data pipeline development across diverse domains.
- Expertise in leveraging Machine Learning (ML) algorithms, including Regression, Classification, Clustering, and segmentation analyses.
- Proficient in Artificial Intelligence (AI) & Deep Learning techniques such as Large Language Model's (LLM), Convolutional Neural Network (CNN) for Computer Vision, Recurrent Neural Network (RNN), with applications of Backpropagation, Stochastic Gradient Descent (SGD), Long Short-Term Memory (LSTM) and text analytics.
- Extensive hands-on experience in Natural Language Processing (NLP) and Generative AI, with a strong foundation in developing Large Language Model (LLM)-based solutions using GPT-4, Gemini, Llama, and Cortex Analyst.
- Deep expertise across the data science project lifecycle from data collection, cleaning, feature engineering, model development/validation to deployment and monitoring.
- Developed forecasting, classification, recommendation and NLP systems leveraging Python, SQL, R, PySpark, TensorFlow, Scikit-learn, and deep learning frameworks, driving measurable business impact.
- Skilled in Python packages such as Pandas, NumPy, SciPy, Scikit-learn, TensorFlow, Matplotlib, Seaborn for data analysis, preprocessing, and machine learning development.
- Proficient in Machine Learning algorithms, XGBoost, Decision Trees, Random Forests, Sentiment Analysis, Naive Bayes Classifier, SVM, and ensemble models.
- Expertise in building Supervised and Unsupervised Machine Learning experiments using Microsoft Azure utilizing multiple algorithms to perform detailed predictive analytics and building Web Services models.
- Implemented data orchestration pipelines using Apache Oozie and managed Hadoop jobs for large-scale machine learning data processing.
- Hands on experience with R packages ggplot2, Shiny, h2o, dplyr, reshape2, Plotly, ElmStatLearn, caTools etc.
- Engineered scalable data pipelines, ETL workflows using Hadoop (MapReduce, Hive) and Spark-based applications (PySpark, Spark SQL) in Databricks to accelerate feature preparation for ML workflows and support business intelligence.
- Integrated Apache Airflow with AWS SageMaker for automated ML lifecycle management including model training and deployment.
- Integrated NLP workflows, using APIs and real-time streaming tools like Kafka and Flume.
- Hands-on with Apache Airflow and AWS Step Functions for workflow orchestration, enabling automated data ingestion, model training, and data publishing.
- Containerized ML services using Docker and Kubernetes to ensure scalable, reproducible, and reliable model deployment.
- Applied CI/CD automation to ML model retraining and deployment pipelines using Jenkins, GitLab CI, and AWS CodePipeline.
- Designed real-time ML pipelines using AWS Lambda, Step Functions, and Kinesis for instant predictions.
- Skilled in designing and implementing robust and scalable data pipelines using Azure Data Factory (ADF), Azure Synapse Analytics, Databricks, AWS Redshift and AutoML to ingest, transform, and deliver data for analytics and machine learning applications.
- Developed advanced SQL queries to create aggregated features and analytical datasets for ML models.
- Experience in creating impactful data visualizations using Python, R and SAS and developed dashboards using tools like Tableau and Power BI.
- Experienced with relational, NoSQL, and columnar databases (Oracle, MySQL, PostgreSQL, Cassandra, MongoDB) to manage diverse ML data requirements.
- Utilized AWS ECS/EKS for scalable ML inference services.
- Experience with Azure DevOps and Git for source control and automated deployment, ensuring reliable and repeatable data and analytics releases.
- Collaborated with cross-functional teams including data engineers, analysts, and business stakeholders to design performant data models and predictive systems.

TECHNICAL SKILLS

Languages: Python (Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch, spaCy), SQL, R, SAS, C/C++, Java, UNIX

Machine Learning & AI: Regression, Classification, Clustering, Deep Learning, XGBoost, LightGBM, CatBoost, SVM, Random Forest, Decision Trees, Prophet, ARIMA, GARCH, Recommender Systems, NLP, Computer Vision

LLMs: GPT-4, Gemini, Llama, RAG, LangChain, Cortex Analyst, Hugging Face Transformers

Databases & Data Warehouses: Oracle, MySQL, PostgreSQL, MS SQL, MongoDB, Cassandra, Redshift, Snowflake, Cloudera

Cloud Platforms: AWS (S3, SageMaker, Step Functions, Lambda, ECS/EKS, Kinesis), Microsoft Azure (Databricks, Synapse, Data Factory, Azure ML), GCP (AI Platform, AutoML, BigQuery)

Big Data Technologies: Hadoop, MapReduce, Kafka, Hive, Spark (PySpark, Spark SQL), Apache Oozie, Airflow, Flume

MLOps & Deployment: MLflow, Kubeflow, Seldon Core, Docker, Kubernetes, Jenkins, GitLab CI, AWS CodePipeline

Visualization & BI Tools: Power BI, Tableau, Streamlit, Matplotlib, Seaborn, ggplot2, Plotly, QlikView

Database Modelling: ER Modelling, Dimension Modelling, Star Schema Modelling, Snowflake Modelling

Tools & Frameworks: Jupyter Notebook, VS Code, PyCharm, Git, Flask, FastAPI

PROFESSIONAL EXPERIENCE

Citi, New York

May 2024 - Present

Sr Data Scientist

Responsibilities:

- Designed and developed end-to-end AI driven predictive analytics solutions for risk assessment, fraud detection, and customer segmentation initiatives.
- Conducted analysis on assessing customer consuming behaviors and discover value of customers with RMF analysis using K-Means Clustering and Hierarchical Clustering.
- Built OCR-based ingestion workflows using OpenCV, combined with spaCy/NLTK for text extraction, named entity recognition (NER), and document classification.
- Performed feature engineering, data cleaning, and dimensionality reduction on structured and semi-structured data using Python (Pandas, NumPy, SciPy) and PySpark.
- Experimented with TensorFlow and Keras for deep learning models focusing on customer churn prediction and unstructured document classification tasks.
- Built regression models including Lasso, Ridge, SVR, XGBoost to predict Customer Life Time Value and classification models Logistic Regression, SVM, Decision Tree, Random Forest to predict Customer Churn Rate.
- Integrated FAISS and Weaviate vector databases for semantic search, enabling efficient retrieval of contextually relevant financial information.
- Led the transition to LLM-powered solutions, designing and deploying a clinical document intelligence system utilizing GPT-4 APIs, LangChain, and Retrieval-Augmented Generation (RAG) for enhanced summarization and decision support.
- Engineered scalable data pipelines on Databricks using PySpark and Spark SQL to extract, transform, and aggregate large-scale banking datasets.
- Developed streaming and batch ETL pipelines with Kafka and Apache Airflow, enabling real-time and scheduled data integration from various banking systems.
- Deployed ML solutions using AWS (S3 for storage, EC2 for compute, Glue for ETL) and explored model deployment alternatives with Azure ML Studio.
- Utilized AWS EKS/ECS to containerize and deploy machine learning inference services for production workloads.
- Orchestrated machine learning pipelines using Apache Oozie, ensuring robust job scheduling in Hadoop environments.
- Applied rigorous statistical analysis (t-tests, Chi-square, ANOVA) and hypothesis testing (A/B) to validate model assumptions and ensure predictive accuracy.
- Created data visualizations using Matplotlib and Seaborn for model diagnostics, including confusion matrices, ROC curves, and feature importance plots.
- Designed and maintained business intelligence dashboards in Tableau and Power BI, visualizing key KPIs for credit risk, churn, and product performance.
- Performed sentiment analysis on customer feedback using NLP to improve customer engagement strategies and deployed to AWS EMR cluster.

- Managed deployment of machine learning solutions across cloud platforms including AWS, Azure, and GCP to ensure scalability and reliability.
- Utilized Azure Synapse Analytics for real-time KPI reporting dashboards to support management decision-making.
- Containerized LLM workflows using Docker, with deployments across AWS SageMaker and Azure ML Studio to ensure scalable and secure access to AI services.
- Participated in risk model governance reviews, preparing documentation and model artifacts for internal audit and compliance verification.

Environment: Python, SQL, R, PySpark, ETL, Hadoop, Hive, Power BI, Tableau, GPT4, LangChain, RAG, FAISS, FastAPI, Machine Learning (Regression, SVM, Decision Tree, Random Forest, XGBoost), NLP, Computer Vision, Databricks, Kafka, Hadoop, Airflow, AWS (S3, EC2, SageMaker, EMR, ECS/EKS), Azure (ML Studio, Synapse), GCP AI Platform, Docker, Git, Jenkins

Elevance Health, Indianapolis
Sr Data Scientist

Jan 2023 - Apr 2024

Responsibilities:

- Designed and implemented predictive analytics solutions to improve patient care outcomes, reduce readmission rates, and optimize healthcare delivery processes.
- Built deep learning architectures (LSTM, CNN, RNN) using TensorFlow to analyze clinical notes and imaging metadata.
- Extracted text and numerical features from documents using Python libraries like BeautifulSoup and created predictive models, including Decision Trees, to classify patient data and predict disease outcomes.
- Conducted exploratory data analysis to identify regional health trends, cost drivers, and care outcome disparities across diverse patient populations.
- Engineered prompt templates and optimized Large Language Models (LLM) responses through advanced prompt engineering strategies tailored for healthcare-specific use cases.
- Implemented scalable data pipelines on AWS using S3, Glue, and Athena for structured and semi-structured healthcare datasets, enabling efficient querying and model input preparation.
- Developed machine learning pipelines to automate the classification and summarization of unstructured healthcare documents using Scikit-learn, TensorFlow, and spaCy, streamlining manual review processes.
- Automated ingestion and real-time processing of clinical documents from hospital systems using Kafka and PySpark, orchestrated through Apache Airflow.
- Implemented scalable algorithms in Python, SQL, Hadoop, MapReduce, MongoDB, and R, ensuring efficient and scalable healthcare data processing and analysis workflows.
- Performed NLP-based sentiment analysis and entity recognition on patient surveys, provider notes, and call center transcripts to improve patient satisfaction.
- Developed recommendation engines to personalize treatment plans and preventive care initiatives based on patient demographics, clinical history, and risk factors.
- Created and maintained interactive dashboards using Power BI and Tableau for monitoring key healthcare metrics such as readmission rates, claims processing and provider performance metrics.
- Conducted A/B testing to evaluate the impact of Machine Learning driven clinical insights, promoting data-driven decision-making within care teams.
- Applied NLP methods such as word2vec to detect near-duplicate patient records and medical reports, develop machine learning models like label spreading and clustering to group medical data.
- Fine-tuned domain-specific Transformer models (e.g., BioBERT, ClinicalBERT) using Hugging Face Transformers, enhancing performance on specialized clinical text tasks.
- Optimized Spark SQL queries to improve data processing speed using Databricks and Azure Synapse environments.
- Containerized ML workflows using Docker, with deployments across AWS SageMaker and Azure ML Studio for scalable and secure access.
- Delivered visual insights and model results via Tableau dashboards and exposed API endpoints using FastAPI for seamless integration with internal tools.
- Automated model retraining workflows using Apache Airflow, enabling continuous integration of new data and adaptive learning for deployed ML models.
- Validated model performance using healthcare-specific performance metrics such as AUC-ROC, F1-score, Precision-Recall.

- Developed chronic care management data solutions by segmenting at-risk patient groups for targeted intervention and outreach programs.

Environment: Python, SQL, R, PySpark, Hugging Face Transformers, MLlib, Spark SQL, ETL, Hadoop, Hive, MapReduce, Power BI, Tableau, Machine Learning (Regression, Decision Tree, Random Forest, Naive Bayes, SVM, Ensemble Models, CNN, RNN, LSTM, Time Series Forecasting, Recommendation Systems), NLP, Kafka, Apache Airflow, AWS (S3, Glue, Athena, SageMaker), Azure (Synapse, ML Studio, Databricks) Docker, Git, FastAPI, MongoDB

Charter Communications, Stamford

Sep 2021 - Dec 2022

Data Scientist

Responsibilities:

- Implemented machine learning algorithms including Random Forests, SVM, Naive Bayes, and Ensemble models to forecast customer lifetime value and optimize marketing campaigns.
- Engineered scalable big data solutions using Hadoop and Spark to process and analyze large-scale telecom datasets, enabling real-time insights for fraud detection, customer behavior analysis, and network optimization.
- Developed predictive models to proactively identify customer churn and improve retention strategies.
- Automated end-to-end machine learning workflows using Apache Airflow integrated with AWS SageMaker for real-time model training, deployment, and monitoring.
- Built and maintained ETL workflows using Apache Airflow and Python, integrating data from Hadoop HDFS for scalable analytics and machine learning pipelines.
- Engineered features from structured and semi-structured data to enhance predictive model performance on customer segmentation and usage forecasting.
- Developed real-time fraud detection models to identify unusual calling patterns and safeguard network integrity.
- Conducted sentiment analysis on customer feedback and social media posts and service call transcripts using NLP techniques to support service quality improvements.
- Created time series forecasting models to predict network bandwidth demand, usage spikes, and maintenance schedules for proactive capacity management.
- Designed interactive dashboards with Power BI and Tableau, visualizing key telecom KPIs such as churn rates, network latency, and customer satisfaction scores.
- Integrated diverse datasets including call detail records (CDRs), customer service logs, and CRM datasets to improve customer experience analytics.
- Performed exploratory data analysis using Python to identify network bottlenecks, quality of service issues, and patterns in customer usage behavior.
- Built anomaly detection models to alert network faults, equipment failures, and potential service outages before customer impact.
- Collaborated with network engineers, data scientists, analysts, and marketing teams to deliver end-to-end analytics and data-driven solutions.
- Validated the performance of predictive models using telecom-specific metrics along with standard evaluation metrics like ROC/AUC, MAE, RMSE, and F1-score.
- Supported deployment of scalable ML inference pipelines on AWS ECS/EKS to enable edge computing for network analytics and predictive maintenance.
- Participated in Agile development cycles, providing data science expertise during sprint planning, reviews, and retrospectives.
- Mentored junior data scientists and analysts on data science practices, machine learning methods, and cloud infrastructure usage.
- Maintained version control, documentation, and audit trails to ensure reproducibility and compliance with industry standards.

Environment: Python, SQL, Spark SQL, ETL, Hadoop, Power BI, Tableau, Machine Learning (Regressions, SVM, Naive Bayes, Decision Tree, Random Forest, Ensemble Models, RNN, LSTM, Time Series Forecasting), NLP, Kafka, Airflow, AWS (Step Functions, SageMaker, ECS/EKS), Azure, GCP, Agile, Docker, Git

Responsibilities:

- Developed machine learning pipelines to automate the classification and summarization of unstructured healthcare documents using Scikit-learn and TensorFlow, streamlining manual review processes.
- Built predictive models for identifying high-priority patient cases using logistic regression, naive bayes and random forest, supporting clinical risk assessments.
- Conducted advanced qualitative and quantitative analysis of high-volume databases to identify trends, patterns, and correlations, driving improved business performance.
- Developed data analytical solutions to analyze results, generating actionable insights and recommendations for company-wide use.
- Designed and maintained data validation and quality assurance workflows using Pandas, NumPy, and custom Python scripts, ensuring integrity across Machine Learning pipelines.
- Automated model retraining workflows using Apache Airflow, enabling continuous integration of new data and adaptive learning for deployed ML models, while fostering a collaborative and solutions-oriented team environment.
- Delivered end-to-end data science solutions involving data collection, cleaning, feature engineering, model development, validation, and deployment to solve complex business problems.
- Built and maintained scalable ETL pipelines on Databricks using PySpark and Spark SQL to process POS, inventory, and customer interaction data.
- Designed and implemented data processing workflows with GCP services such as Google Cloud Dataproc, Dataflow, and Dataprep, ensuring efficient and scalable data transformations.
- Utilized GCP's Pub/Sub and Dataflow to implement real-time data streaming and processing solutions, providing timely insights and analysis from streaming data sources.
- Leveraged Hadoop, MapReduce, Hive, Spark Streaming, and Snowflake for large-scale retail sales analytics and demand forecasting.
- Integrated and transformed data from diverse sources using GCP's data integration tools, maintaining data quality and consistency through effective ETL processes.
- Worked with product owners, business analysts, and infrastructure engineers to design and build scalable, fault-tolerant applications on Kubernetes following agile methodologies.
- Partnered with stakeholders to understand data requirements and develop dashboards, data visualizations, and business case analyses in Power BI.
- Collaborated within a data engineering team to gather requirements and perform business analysis, translating business needs into data pipeline designs and implementations on GCP using BigQuery, Python, Shell, Airflow, GitHub and Jenkins.

Environment: Python, SQL, Machine Learning (Logistic Regression, Naive Bayes, Random Forest), PySpark, Spark SQL, Hadoop, Hive, MapReduce, BigQuery, Pub/Sub, Cloud Storage, Cloud Composer, Airflow, Databricks, Data Science, Spark, Kubernetes, Snowflake, Power BI, GitHub, Jenkins